

Grundfos CR45-3 Vertical Multistage Pump — Repair Manual

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EQUIPMENT

Tag: EQ-PMP-CR45-01 Plant: South Plant Utilities
Model: CR45-3-2 A-F-A-E-HQQE Category: Vertical Multistage Pump
Serial: GF-CR45-30022
Manufacturer: Grundfos (Denmark)

SAFETY HAZARDS

- High-pressure water
- Pinch points
- Suspended load during lift

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Section 3 — Stage Disassembly

Procedure ID: PRC-PMP-005-A

Category: Corrective Maintenance Severity: Medium

Performed by: T-1001 Jordan Reyes (Rotating Equipment)

OVERVIEW

This procedure covers replace stage chamber gaskets on equipment EQ-PMP-CR45-01 (CR45-3-2 A-F-A-E-HQQE). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED

- GSKT-ANSI-150 — ANSI 150 Spiral-Wound Gasket DN50 (Gaskets)
- SEAL-MS-25-A — Mechanical Shaft Seal 25mm Type A (Seals)

STEPS

1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 3 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA

- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.

Section 4 — Belt Service

Procedure ID: PRC-PMP-005-B

Category: Preventive Maintenance Severity: Low

Performed by: T-1208 Aisha Kone (Reliability Engineering)

OVERVIEW

This procedure covers v-belt drive replacement on equipment EQ-PMP-CR45-01 (CR45-3-2 A-F-A-E-HQQE). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED

- BELT-V-A56 — Classic V-Belt A56 (Drives)

STEPS

1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 4 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA

- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.